

AI Engineering Basics

Session 1: What is AI? and Understanding data

Concept-only: no maths, no coding

Click, explore and try it yourself

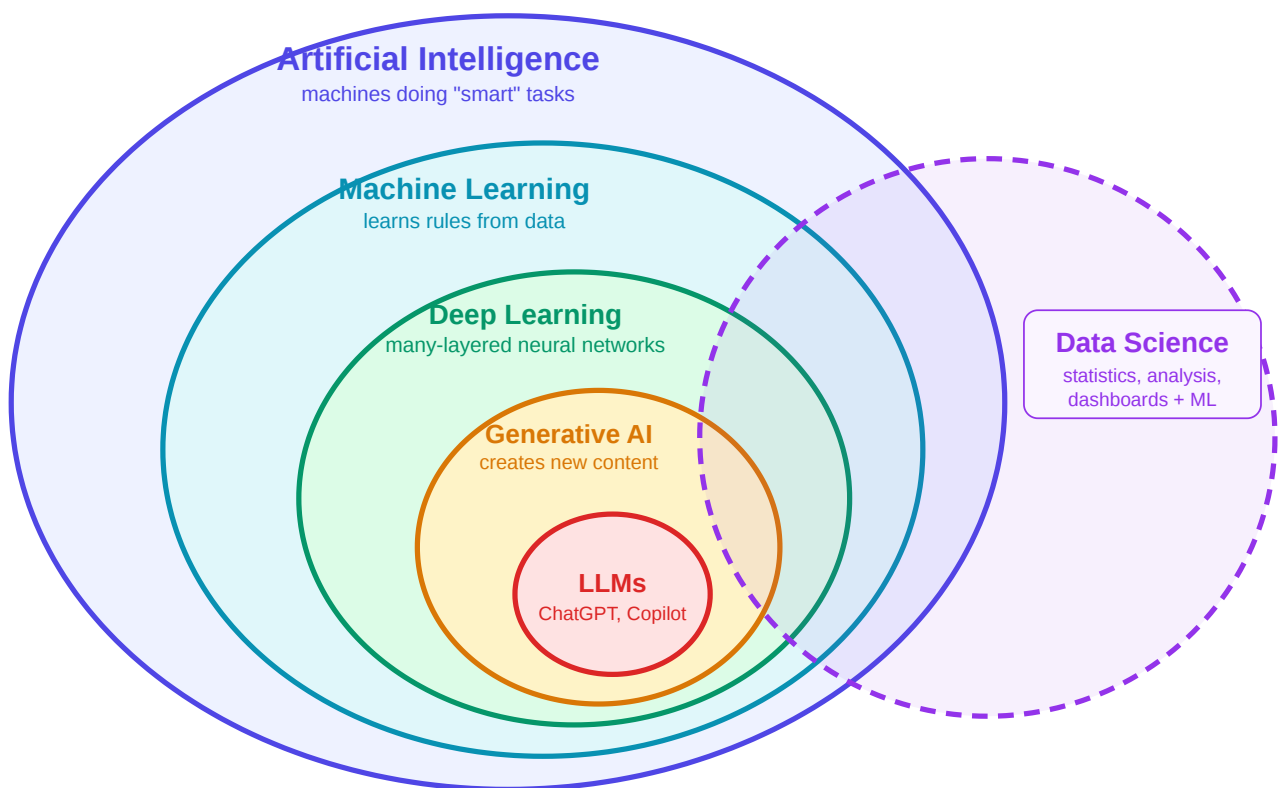
Includes a dataset cheat sheet

THE STORY OF THIS SESSION

What is AI? → what does it learn from (data)? → what words do we use for data? → learning **with** answers (supervised) and **without** answers (unsupervised) → how do we sample and split the data fairly? → how do we know the model really learned (overfitting, underfitting, generalisation)? **Next session** opens the model and shows *how* it learns.

1. The AI family: how everything fits together

AI, machine learning, deep learning, generative AI and LLMs sit **inside each other**. Data science is a **separate field** that **overlaps** with them. It is not the parent of AI.



Start here

Nested circles: each one is a special kind of the circle around it

1. **AI**: the goal: smart machines
2. **Machine learning**: AI that learns from data
3. **Deep learning**: ML with many layers
4. **Generative AI**: DL that creates
5. **LLMs**: GenAI for language

Data science (dashed circle) is a **neighbouring field**. It overlaps with ML and DL, but it doesn't contain AI.

Artificial Intelligence

The biggest circle: the goal

Any technique that makes a machine do something we would call "intelligent": deciding, recognising, planning, answering. AI is the **goal**, not one technique.

Early AI was **rule-based**: humans wrote every IF/THEN rule by hand. Modern AI mostly **learns from data** instead (that's ML, the next circle).

ANALOGY

A recipe book. The machine follows rules someone else wrote down.

Examples

- ✓ Rule-based chess engine
- ✓ Traffic-light controller
- ✓ IF/THEN medical expert system
- ✓ Robot path planning

Not an example

- ✗ Sales dashboard (data science, not AI)

Machine Learning

Inside AI: learning from examples

Instead of writing the rules, we **show the machine many examples** and it **works out the rules itself**.

Traditional programming: **rules + data** → **answers**.

Machine learning: **data + answers** → **rules**.

Three styles: **supervised** (with answers), **unsupervised** (finds groups), **reinforcement** (trial and reward).

ANALOGY

Teaching a child what a "dog" is by showing many dogs, not by giving a definition.

Examples

- ✓ ML spam filter
- ✓ Loan approval prediction
- ✓ Customer grouping
- ✓ Netflix-style recommendations

Not an example

- ✗ Hand-written IF/THEN rules (AI, but not ML)

Deep Learning

Inside ML: many layers

Machine learning using **neural networks with many layers**, loosely inspired by the brain. "Deep" just means **many layers**.

Classic ML often needs humans to decide *what to look at*. Deep learning **finds the useful features itself**: early layers see edges, middle layers see eyes and ears, final layers say "cat".

ANALOGY

A factory assembly line. Each station adds a little more understanding.

Examples

- ✓ Face unlock
- ✓ Voice assistants
- ✓ Google Translate
- ✓ Medical image scanning

Not an example

- ✗ Simple loan model on a small Excel sheet (usually classic ML)

Generative AI

Inside deep learning: creates new content

Deep learning models that **create new content** (text, images, code, music, video) instead of only labelling or predicting.

Predictive AI answers "**which one / how much?**". Generative AI answers "**make me one**".

ANALOGY

A spam filter is a **judge**. Generative AI is an **author**.

Examples

✓ DALL·E / Midjourney (images)

✓ Music and video generators

✓ ChatGPT, Copilot (text)

Not an example

✗ Face unlock (recognises, doesn't create)

Large Language Models (LLMs)

Inside generative AI: models for language

Very large generative models trained on huge amounts of text. At their core they do one thing: **predict the next word (token)**, again and again.

They are built on the **Transformer** architecture (2017). We'll reach this later in the course.

ANALOGY

Super-powered autocomplete that has read a large part of the internet.

Examples

✓ ChatGPT

✓ Microsoft Copilot

✓ Claude

✓ Gemini

Not an example

✗ Midjourney (generative AI for images, not language)

Data Science

A separate field that overlaps with AI

The field of **getting insights and decisions from data** using statistics, analysis, visualisation and machine learning.

It is **not the parent of AI**. It overlaps with ML and deep learning because data scientists often use them. Part of data science has nothing to do with AI, and part of AI has nothing to do with data science.

ANALOGY

A detective who uses data as clues: sometimes with AI tools, sometimes just a calculator and charts.

Examples

- ✓ Sales dashboards
- ✓ A/B test analysis
- ✓ Churn prediction with ML (overlap)
- ✓ Business reports

Not an example

- ✗ Rule-based chess engine (AI, not data science)

Why AI is the biggest circle: the subset test

A is inside B only if **every A is also a B**.

Question	Answer	So...
Is every ML system an AI system?	✓ Yes	ML is inside AI
Is every deep learning model an ML model?	✓ Yes	DL is inside ML
Is every generative AI model a deep learning model?	✓ Yes	GenAI is inside DL
Is every LLM a generative AI model?	✓ Yes	LLMs are inside GenAI
Is every AI system data science?	✗ No. A rule-based chess engine uses no data.	AI is not inside data science
Is all data science AI?	✗ No. A sales dashboard is not AI.	Data science is not inside AI

IT ALSO FOLLOWS HISTORY

AI started first (1956) as the **goal** of smart machines. Machine learning (1959) was a way to reach it by learning from data. Deep learning is a kind of ML, generative AI is a kind of deep learning, and LLMs are generative AI for language. "Data science" became popular much later (2000s) as the practice of getting insights from data, and it **borrow**s ML as a tool. That shared ML is the overlap.

One problem, every level: the email example

Level	How it handles the email task
Rule-based AI	A human writes rules: IF the email contains "lottery" → spam. <i>Problem:</i> spammers write "l0ttery" and the rule misses it. Someone has to keep writing rules forever.
Machine learning	Give the model 100,000 emails already marked spam / not spam. It learns the patterns itself (odd senders, many links, ALL CAPS), including tricks nobody wrote a rule for.
Deep learning	It understands the meaning of the whole email, so it catches cleverly worded scams that avoid obvious spam words.
Generative AI	It goes beyond judging emails: it creates things, such as a summary of your inbox or a draft reply.
LLM	Copilot in Outlook: "Summarise this thread and draft a polite reply". It predicts the best next words, one token at a time.
Data science	An analyst builds a dashboard: "How many spam emails did we get per month, and from which countries?" Insight from data; no AI needed.

Predictive AI vs. generative AI

Predictive AI: "which one / how much?"	Generative AI: "make me one"
"Is this email spam?" → Yes	"Write a polite reply to this email" → <i>a new paragraph</i>
"Is there a cat in this photo?" → Yes	"Draw a cat wearing sunglasses" → <i>a new image</i>
"What will this house sell for?" → ₹85 lakh	"Write a listing ad for this house" → <i>new text</i>

Takeaway: AI is the **goal** → ML **learns from data** → deep learning uses **many layers** → generative AI **creates** → LLMs create **language**. Data science **overlaps** through ML but is its own field.

2. Sort it: where does it belong?

Pick the **smallest circle** each example fits into. Good for class voting.

Example	Smallest circle	Why
A 1990s chess program that follows hand-written rules	AI (rules)	Humans wrote the rules and it doesn't learn from data, so it's AI but not ML.
A bank model that learns from past loans who is likely to repay	Machine learning	It learns patterns from labelled past data, usually with classic ML.
Face unlock on your phone	Deep learning	Recognising faces from raw pixels needs many-layered neural networks.
Midjourney drawing a picture from a sentence	Generative AI	It creates a new image: generative AI. Not an LLM, because it makes images, not language.
ChatGPT writing an email	LLM	A large language model predicting the next words.
A monthly sales dashboard in Power BI	Data science (not AI)	Insight from data with charts, and no learning model: data science, not AI.
Netflix suggesting your next show	Machine learning	It learns your preferences from viewing data. (Large systems may also use deep learning.)
Google Translate turning Kannada into English	Deep learning	Modern translation uses deep neural networks, often Transformers.
An A/B test report comparing two website designs	Data science (not AI)	Statistics on data to make a decision: data science, not AI.
Microsoft Copilot summarising a Teams meeting	LLM	It generates language, so it is an LLM.
A thermostat: IF temperature > 26°C THEN switch on AC	AI (rules)	A simple fixed rule: the most basic kind of rule-based automation.
Grouping customers into types with no predefined groups	Machine learning	Unsupervised machine learning: it finds the groups itself.

3. A short history

When	What happened
1950s	The idea of "thinking machines" begins, and the term Artificial Intelligence is coined (1956).
1970s–80s	Rule-based expert systems become popular, with humans writing thousands of IF/THEN rules.
1990s–2000s	Machine learning grows: spam filters, recommendations. IBM Deep Blue beats the world chess champion (1997).
2012	Deep learning breakthrough in image recognition, thanks to big data and GPUs.
2017	The Transformer is invented, the architecture behind today's LLMs.
2022 onwards	ChatGPT launches and generative AI and LLMs go mainstream.

WHY DEEP LEARNING TOOK OFF ONLY AROUND 2012

The ideas were decades old. Three things finally came together: **big data** (the internet), **powerful hardware** (GPUs, originally made for gaming) and **better training techniques**.

4. Datasets: the model's textbook (and the words we use)

From machine learning onwards, everything **learns from examples**. That collection of examples is a **dataset**.

ANALOGY

If the model is a **student**, the dataset is the **textbook**. A brilliant student with a wrong textbook still learns the wrong things.

Anatomy of a dataset

Size (sq ft)	Bedrooms	Location	Age (yrs)	Price (label)
1,200	2	Whitefield	5	₹75 L
1,800	3	Indiranagar	10	₹1.6 Cr
950	2	Electronic City	2	₹52 L
2,400	4	Koramangala	8	₹2.3 Cr

Same thing, different names

Books, videos and colleagues switch between these names freely. Learn the groups once and you'll never be confused.

Idea	Other names you'll hear	Symbol	In the house table
Feature	input, independent variable , predictor, attribute, column, explanatory variable	x / X	Size, bedrooms, location, age
Label	target , dependent variable , output, response, ground truth, class (in classification)	y	Price
Row	record , sample, example, instance, observation, data point	n = number of rows	One house
Numerical feature	continuous or count values: amounts		Size, age
Categorical feature	names or groups		Location

! CAREFUL: X IS NOT THE CLASS

X is always the **inputs** (features). The **class** is a name for the **answer** y in a classification problem (for example "spam" / "not spam"). So: X = features, y = label / target / class.

WHY "INDEPENDENT" AND "DEPENDENT"?

The price **depends on** the size, so price is the **dependent** variable. The size doesn't depend on the price, so it's **independent**. Features are what we feed in; the label is the result we want to predict.

Quick check: which is it?

Spam dataset: the column "number of links in the email"

Feature (X)

Label (y)

Row

An input the model looks at: a feature.

Spam dataset: the column "is_spam (yes/no)"

Feature (X)

Label (y)

Row

The answer: the label, also called the class here.

One single email with all its columns

Feature (X)

Label (y)

Row

One record / sample / row.

"Dependent variable" in a salary-prediction dataset

Feature (X)

Label (y)

Row

Salary depends on the inputs, so it's the target / label.

"Predictor" or "independent variable"

Feature (X)

Label (y)

Row

Other names for a feature.

"Observation" or "instance"

Feature (X)

Label (y)

Row

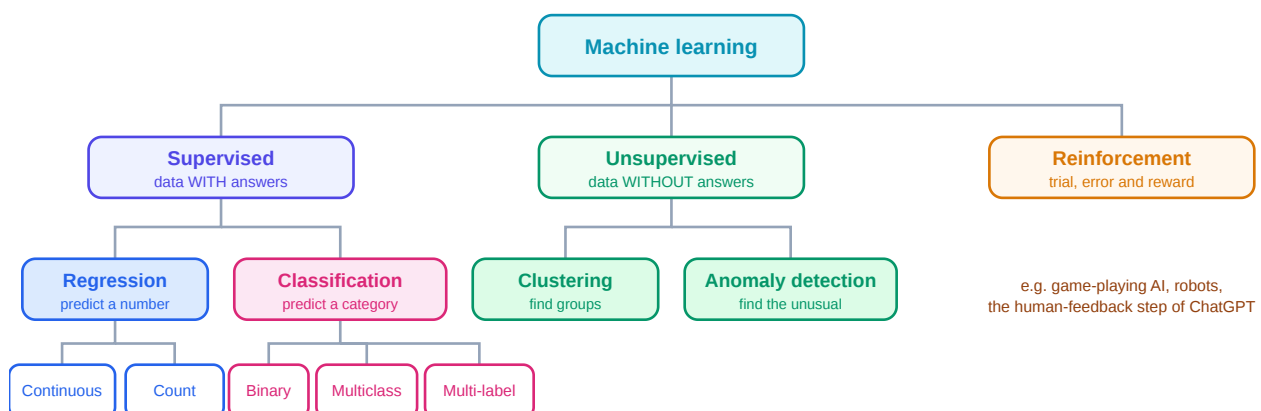
Other names for a row.

Why datasets matter

- **They are the only teacher.** The model knows nothing except what the data shows it.
- **They set the limits.** A model trained only on Bengaluru houses can't price houses in Delhi.
- **They pass on mistakes.** Wrong or biased data produces a wrong or biased model: **garbage in, garbage out.**
- **They test the model.** Part of the data is kept aside to check whether the model really learned (section 9).

The map of learning tasks

What kind of data you have decides what kind of learning you can do.



THE ONE QUESTION THAT DECIDES IT

Does my data have the answer (label) in it? Yes → **supervised** (section 6). No → **unsupervised** (section 7).

For supervised: is the answer a number or a category? Number → **regression**. Category → **classification**.

Takeaway: Each **row** is one record. **X** = features (independent variables). **y** = label / target / class (dependent variable).

5. Types of datasets and data quality

The same dataset can belong to more than one group.

By structure

Type	What it is	Examples
Structured (tabular)	Neat rows and columns, like Excel	Bank transactions, sales records, house prices
Unstructured	No fixed rows or columns	Emails, reviews, images, audio, video
Semi-structured	Some organisation, but flexible	JSON files, system logs, web pages

By labels

Type	What it is	Example	Used for
Labelled	Each example has its answer	Photos tagged "cat" or "dog"	Supervised learning
Unlabelled	Examples only, no answers	A million customer records with no groups	Unsupervised learning
Partly labelled	A few labelled, most not	100 tagged scans + 10,000 untagged	Semi-supervised learning

WORTH KNOWING

Labelling is often the most expensive part of a project, because humans usually do it by hand (for example, doctors labelling X-rays).

By data type

Data type	Example dataset	Typical problem
Numbers / tables	Loan applications	Will this person repay?
Text	Product reviews	Positive or negative?
Images	Chest X-rays	Any sign of disease?
Audio	Voice recordings	Speech to text
Video	CCTV footage	Detect unusual activity
Time series	Daily electricity use	What will tomorrow look like?

By source

Source	Examples	Good for
Public / open	Kaggle, Hugging Face, UCI, data.gov.in	Learning and experiments
Internal / company	Support tickets, sales data, logs	Real business problems (needs permission)
Collected / scraped	Surveys, sensors, web pages	When nothing suitable exists (check the legal side)
Synthetic	Generated by a program or by AI	When real data is rare, private or expensive

Spot the problems: a messy dataset

Customer ID	Name	City	Age	Monthly spend	Cancelled?
C001	Priya	Bengaluru	29	₹2,400	No
C002	Rahul	Bangalore	34	₹1,900	Yes
C003	Anita	Mumbai		₹3,100	No
C004	Vikram	Chennai	290	₹2,050	No
C001	Priya	Bengaluru	29	₹2,400	No
C006	Meera	Delhi	41	₹1,20,000	No
C007	Arjun	Pune	26	₹2,700	Y

- Inconsistent format: 'Bangalore' and 'Bengaluru' are the same city written differently.
- Missing value: age is blank.
- Outlier: an age of 290 is impossible, probably a typo for 29.
- Duplicate: C001 (Priya) appears twice.
- Unit error: ₹1.2 lakh per month doesn't match the others; possibly yearly spend or a typo.
- Inconsistent label: 'Y' should be 'Yes'. Labels must use one consistent format.

Common data quality problems

Problem	Example
Missing values	Age not filled in for 20% of customers
Duplicates	One customer counted three times
Wrong labels	A dog photo tagged "cat"
Inconsistent format	"Bengaluru", "Bangalore", "BLR"

Outliers	A house with 45 bedrooms
Imbalance	99 normal transactions for every 1 fraud
Bias	A hiring dataset with mostly one gender
Outdated	Shopping data from before online payments were common

WHY IMBALANCE TRICKS US

If 99% of transactions are normal, a lazy model that *always* says "not fraud" is **99% accurate**, and it catches zero fraud.

Takeaway: The dataset is the model's **teacher**. A model is only as good as the data it learns from.

6. Supervised learning: regression vs. classification

In supervised learning, every example has an answer (label). What **kind** of answer it is decides the type of problem.

Regression: "How much? How many?"

The answer is a **number on a scale**. Being "close" matters: predicting ₹74 L when the truth is ₹75 L is a good guess.

Examples: house price, delivery time, tomorrow's temperature, number of tickets tomorrow.

Classification: "Which one? Which group?"

The answer is a **category (a name or tag)**. There's no "close": an email is either spam or not spam.

Examples: spam / not spam, which team gets this ticket, which digit is in an image.

THE 2-SECOND TEST

Ask: "Can I take an average of the answers, and does 'close' make sense?"

✓ Yes → **regression**. ✗ No (you can't average "Billing" and "Login") → **classification**.

Continuous vs. discrete: clearing up a common confusion

Kind of answer	What it means	Examples	Usually solved with
Continuous number	Any value, including decimals	₹74.6 L, 27.3°C, 18.5 minutes	Regression
Discrete count	Whole numbers only, but still an amount	Tickets tomorrow: 0, 1, 2, ... 140	Regression (count regression)
Discrete category	A fixed set of names or groups	Spam / not spam; Billing / Login / Bug	Classification

! TRAP: NUMBERS THAT ARE REALLY CATEGORIES

A PIN code, an employee ID or a product code **look** like numbers but are really **names**, so predicting them is **classification**. **In-between case:** a 1–5 star rating is ordered, so it can be treated either way.

Types of regression (concept level)

Type	Plain meaning	Example
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Simple	One input predicts the number	House size → price
Multiple	Many inputs together predict the number	Size + location + age → price
Linear	Roughly a straight line	Every extra 100 sq ft adds about the same amount
Non-linear	The relationship curves or flattens out	Crop yield rises with fertiliser, then flattens, then drops
Count	Whole-number counts that can't go below zero	Patient visits to a clinic tomorrow

Types of classification

Type	Plain meaning	Example
Binary	Exactly 2 possible answers	Spam / not spam · repay / default
Multiclass	3 or more answers; pick exactly one	Ticket → Billing, Technical, Account or Other
Multi-label	Several answers can be true at once	A movie tagged Action + Comedy

ANALOGY

Binary = a light switch. **Multiclass** = a TV remote (exactly one channel). **Multi-label** = pizza toppings (as many as apply).

How do we judge success? (plain words)

	Regression	Classification
Main question	"On average, how far off are we?"	"How often are we right, and do we catch the important cases?"
Example	"Our price predictions are off by about ₹4 L on average."	"We catch 90 of every 100 frauds; 5 of every 100 alerts are false alarms."

Try it: regression or classification?

Price of a used car

Regression

Classification

Regression: continuous.

Will this customer cancel next month? (yes / no)

Regression

Classification

Classification: binary.

Tomorrow's maximum temperature in Bengaluru

Regression

Classification

Regression: continuous.

Which language is this text in?

Regression

Classification

Classification: multiclass.

How many support tickets will arrive tomorrow?

Regression

Classification

Regression: count.

Is this tumour benign or malignant?

Regression

Classification

Classification: binary.

Which genres does this movie belong to? (can be several)

Regression

Classification

Classification: multi-label.

Food-delivery time in minutes

Regression

Classification

Regression: continuous.

PIN code of the customer's area

Regression

Classification

Trick! It's a name for a place, not an amount.

Number of units of a product sold next week

Regression

Classification

Regression: count.

Worked examples

Regression: continuous

Predict the selling price of a flat in Bengaluru

Output

A price in ₹

Features (X)

Size, bedrooms, location, age, floor

Label (y)

Actual selling price

Size (sq ft)	Beds	Location	Age	Price
1,200	2	Whitefield	5	₹75 L
1,800	3	Indiranagar	10	₹1.6 Cr
950	2	Electronic City	2	₹52 L

How we judge it

"On average we're off by about ₹4 L."

Watch out for

Outliers; prices change over time

Regression: continuous

Predict how many minutes a food delivery will take

Output Minutes
Features (X) Distance, time of day, rain, prep time
Label (y) Actual delivery time

Distance (km)	Time	Raining?	Prep (min)	Minutes
2.1	13:10	No	12	24
5.8	20:45	Yes	18	51
3.4	16:00	No	9	29

How we judge it "On average our estimate is off by 4 minutes."
Watch out for Festival days; GPS errors

Regression: count

Predict how many support tickets arrive tomorrow

Output A whole number ≥ 0
Features (X) Day, holiday?, release yesterday?, 7-day average
Label (y) Actual ticket count

Day	Holiday?	Release yesterday?	7-day avg	Tickets
Mon	No	Yes	88	142
Sat	No	No	90	31
Tue	Yes	No	85	12

How we judge it "On average we're off by about 9 tickets."
Watch out for Time-based data: split by date

Classification: binary

Will this applicant repay or default?

Output Repay / Default

Features (X) Income, loan amount, credit score, existing EMIs

Label (y) What happened with past loans

Income (₹/mo)	Loan (₹)	Credit score	EMIs	Outcome
85,000	10 L	780	1	Repaid
32,000	8 L	610	3	Defaulted
60,000	5 L	720	0	Repaid

How we judge it "We catch 85 of every 100 defaulters, and wrongly reject 6 of every 100 good applicants."

Watch out for Imbalance; bias; privacy

Classification: multiclass

Route each support ticket to the right team

Output Exactly one of Billing / Technical / Account / Other

Features (X) Ticket subject and description

Label (y) The team that resolved it

Subject	Description (short)	Team
Charged twice	My card was debited two times...	Billing
App crashes	Crashes when I upload a file...	Technical
Change email	I want to update my login email...	Account

How we judge it "92% of tickets go to the right team first time."

Watch out for Mis-routed past tickets; rare categories

Classification: multi-label

Tag a news article with all topics that apply

Output Any combination of Politics / Business / Tech / Sports

Features (X) Headline and text

Label (y) Tags assigned by editors

Headline	Tags
Govt announces new AI policy for startups	Politics, Tech, Business
Cricket board signs ₹5,000 Cr media deal	Sports, Business
New smartphone launches in India	Tech

How we judge it Per tag: how many correct tags found, how many wrong tags added.

Watch out for Inconsistent tagging; rare tags

Takeaway: A **number on a scale** → regression (continuous or count). A **category** → classification (binary, multiclass or multi-label).

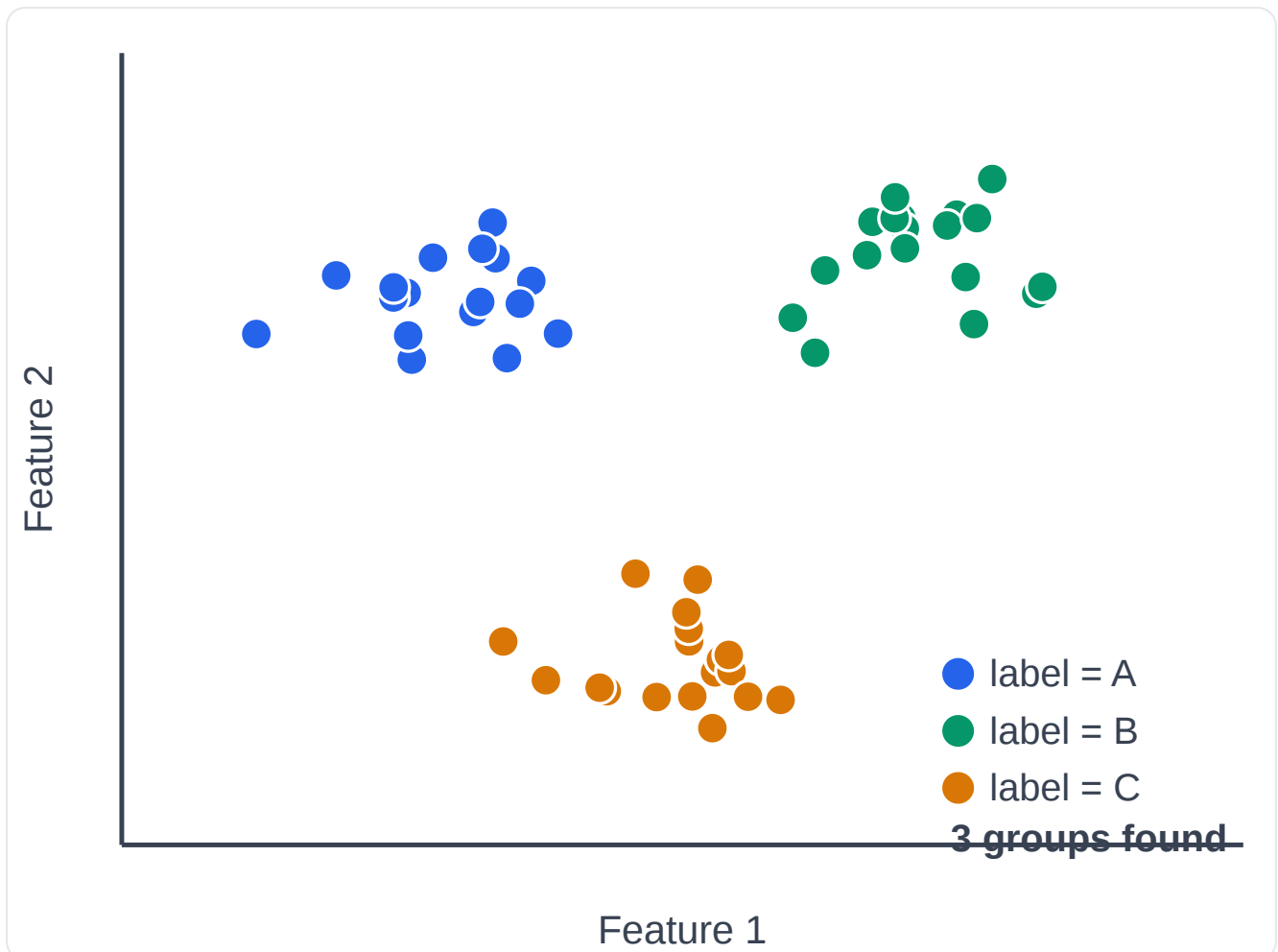
7. Unsupervised learning: when there are no labels

So far every dataset had an answer column (y). But very often we have **only the features (X)** and no answers at all. Can a machine still learn something? **Yes**. It can discover **structure** by itself: which rows are similar, which rows are strange.

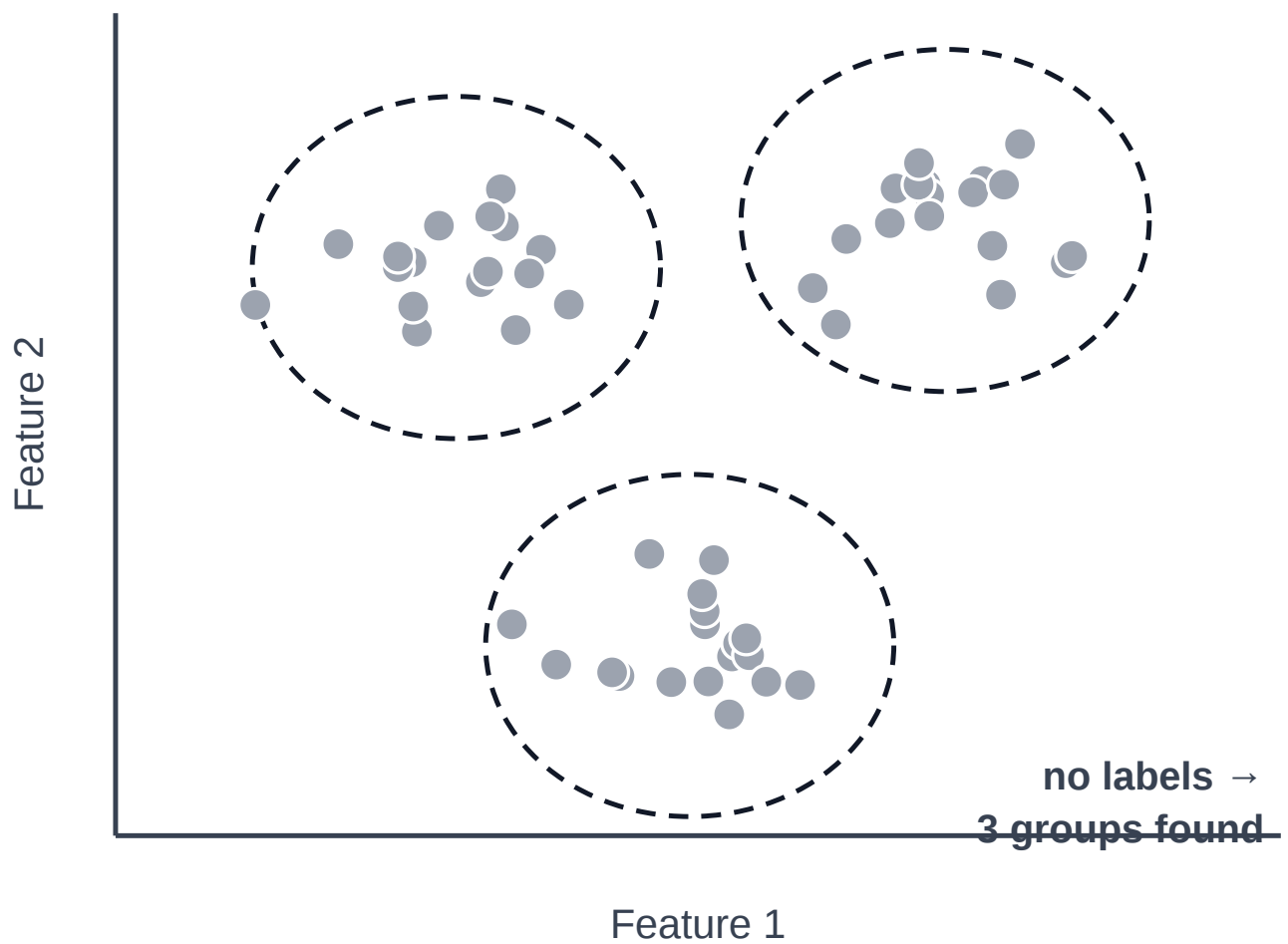
ANALOGY: SORTING A MIXED BOX OF LEGO

Nobody gives you instructions or names for the pieces. You still naturally put similar pieces together: by colour, by size, by shape. You **found the groups yourself**. That is unsupervised learning.

Same data, two situations

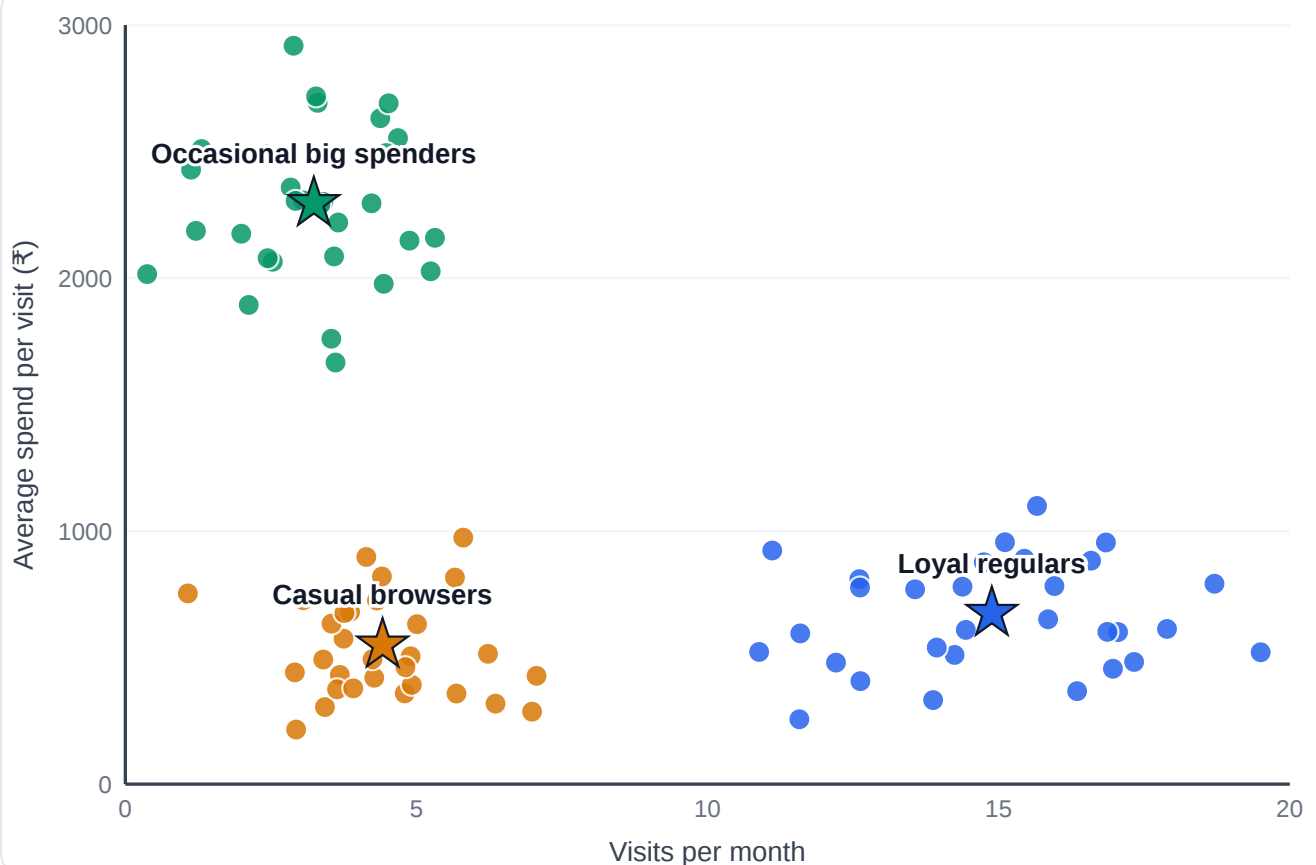


Supervised: every point comes with its answer (colour = label). The model learns to predict the colour of a new point.



Unsupervised: the same points, but **no answers** (all grey). The model notices the points naturally form groups, and draws them.

Try it: watch the machine find customer groups (clustering)



✓ **Finished after 3 rounds:** nothing changes any more. The machine found 3 groups by itself. We (humans) named them from what they look like.

HOW THE MACHINE DOES IT (ONE POPULAR METHOD)

1. Drop a few **centre markers** (the stars), spread across the data.
2. Every customer joins the **nearest** centre.
3. Each centre moves to the **middle** of its group.
4. Repeat 2–3 until nothing changes.

THE MACHINE FINDS GROUPS; HUMANS GIVE THEM MEANING

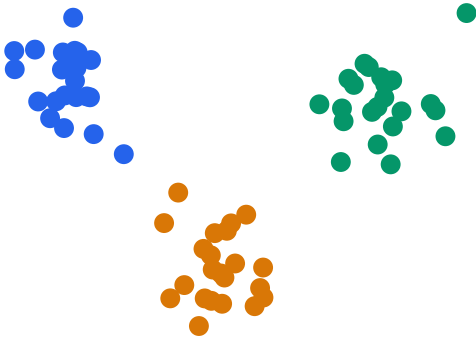
With 3 groups, the machine finds something like this, but it only knows "group 1, 2, 3". **We** look at each group and name it:

- **Occasional big spenders:** few visits, high spend per visit.
- **Loyal regulars:** many visits, small baskets.
- **Casual browsers:** few visits, small spend.

The marketing team can now send a different offer to each group, without anyone ever labelling a customer.

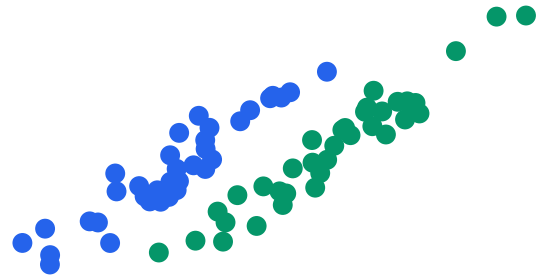
What clusters can look like

Real groups aren't always neat circles. Different methods are good at different shapes, and some methods can also say "this point doesn't belong to any group" (noise).



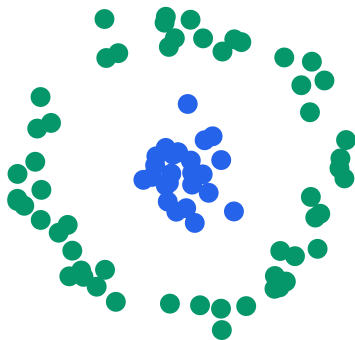
Round, well-separated

The easy case: clear, compact groups.



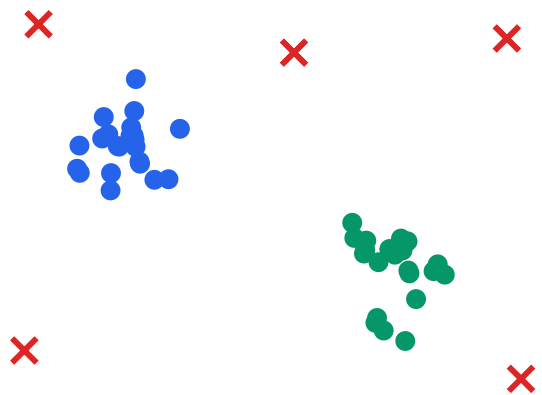
Stretched groups

Long, tilted groups that sit close together.



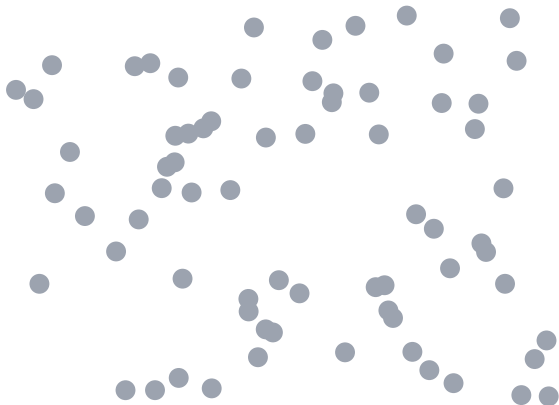
Rings and curved shapes

One group wraps around another. Some methods follow the shape instead of looking for round blobs.



Groups + noise

Some methods mark points that fit nowhere as **noise** (×). Those are candidates for anomaly detection.



No real groups

Evenly spread data. A method will still split it if you ask, so check the groups mean something.

What unsupervised learning is used for

Task	What the machine does	Real examples
Clustering	Puts similar rows into groups nobody defined in advance	Customer segments · grouping similar news articles · grouping support tickets by topic · grouping patients with similar symptoms
Anomaly detection	Learns what "normal" looks like and flags rows that don't fit	Unusual card transactions · a machine sensor behaving strangely · odd login activity
Association	Finds things that often happen together	"People who buy bread often buy butter" · "frequently bought together" on shopping sites
Simplifying data (dimensionality reduction)	Squeezes many columns into a few, keeping the important information	Turning 100 survey questions into 3 main themes · making data easier to plot

Supervised vs. unsupervised at a glance

	Supervised	Unsupervised
Data	X and y (features + answers)	Only X (features)
Goal	Predict the answer for new rows	Discover groups, patterns or odd rows
How we check it	Compare predictions with the real answers	No answers to compare with: we check that groups are tight and well separated, and that they make sense to people who know the business
Example	"Is this transaction fraud?" (using past fraud labels)	"Which transactions look unusual?" (no labels)

! WATCH OUT

The machine will **always** find the number of groups you ask for, even if the data has no real groups. Ask for 5 groups in the demo above and it happily makes 5. Choosing the number of groups, and checking the groups mean something, is **our** job.

Takeaway: No labels? The machine can still learn the **shape** of the data: which rows are similar (clustering), which rows are strange (anomaly detection), and what goes together. Humans then give the groups meaning.

8. Sampling: choosing which rows to use (and the random seed)

Sampling means picking a smaller group (a **sample**) from a bigger group (the **population**). It's the first step to splitting data fairly.

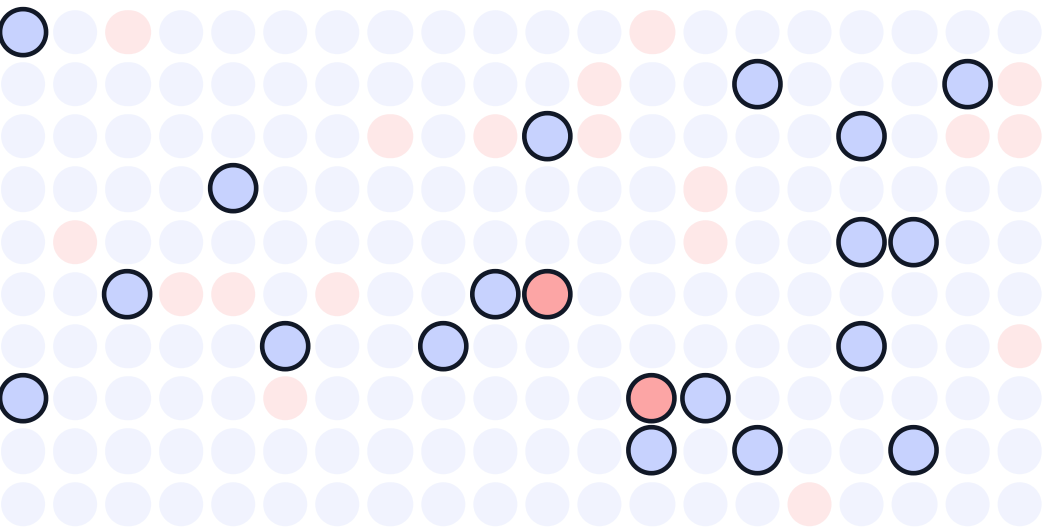
ANALOGY

A cook tastes **one spoon** of soup to judge the whole pot. That only works if the soup is stirred well. A bad sample (only the top layer) gives the wrong answer.

Where sampling shows up in ML

When	What we sample
Collecting data	We can't survey every customer in India, so we survey a sample.
Splitting data	Train / validation / test sets are samples of the full dataset (next section).
Training	Every mini-batch is a small random sample of the training rows (Session 2).
Fixing imbalance	We over-sample the rare class or under-sample the common one.

Try it: random vs. stratified sampling



Stratified (2 from fraud, 18 from normal) (seed 42): picked 20 customers, **2 fraud** (10%). Ideal: 2 (10%). ✓ Matches the population.
First picked rows: #1, #35, #39, #51, #57, #65, ...

Types of sampling

Type	How it works	Example	Good / watch out
------	--------------	---------	------------------

Simple random	Every row has an equal chance	Pick 1,000 random customers	✓ Fair and simple. ! Can miss rare groups by bad luck.
Stratified	Split into groups first, then sample each group in the right proportion	Keep 10% fraud in train <i>and</i> test	✓ Best for imbalanced data and train/test splits.
Systematic	Pick every k-th row	Every 10th invoice	✓ Easy. ! Breaks if the data has a repeating pattern.
Cluster	Pick whole groups, then use all rows in them	Pick 5 cities, survey everyone there	✓ Cheap. ! Chosen groups may not represent the rest.
Convenience	Take whatever is easiest	Survey only your office colleagues	! Usually biased. Avoid for real models.
Over-sampling	Repeat (or create similar) rows of the rare class	Copy fraud rows so the model sees more	✓ Helps imbalance. ! Training data only.
Under-sampling	Drop some rows of the common class	Keep 10% of normal transactions	✓ Balanced, faster. ! Throws information away.

WITH OR WITHOUT REPLACEMENT

Without replacement: once a row is picked, it can't be picked again (normal splitting). **With replacement:** a row can be picked more than once (**bootstrapping**, used later by random forests).

The random seed (random state): making "random" repeatable

Computers don't produce truly random numbers. They follow a long, fixed recipe that only **looks** random. The **seed** is the **starting point** of that recipe. Same seed → the same "random" sequence → the **same sample, the same split, every time**.

ANALOGY: SHUFFLING CARDS WITH A RECORDED SHUFFLE

If you write down exactly how you shuffled a deck, a friend can repeat the shuffle and get the **identical order**. The seed is that written-down shuffle.

Why it matters	Example
Reproducibility	You and a colleague both split the data with seed 42 → you get exactly the same training and test rows, so your results match.
Fair comparisons	Comparing two models? Use the same seed so both see the same split. Otherwise one may just have got an easier test set.

Debugging

If a result looks strange, you can re-run and get the same thing, instead of a different result each time.

In tools you'll see it as `random_state` or `seed` (for example, when splitting data into train and test). The number itself doesn't matter (42 is just a popular joke choice); what matters is **using the same one**.

! DON'T "SHOP" FOR A LUCKY SEED

Trying many seeds and keeping the one with the best test score is cheating yourself: the good score was luck, not a better model.

GOLDEN RULE

Split into train / validation / test **first**. Only then over- or under-sample, and only the **training** set. Otherwise copies of a row can end up in both train and test (data leakage).

Takeaway: A good sample looks like the population in miniature. Use **stratified** sampling when a class is rare, and a fixed **seed / random_state** so anyone can reproduce your exact sample.

9. Train, validation and test: in depth

Why we split the data, why roughly 80% goes to training, and what each part is really for.

THE SCHOOL ANALOGY

Training set = textbook and class notes. The student learns from these.

Validation set = practice / mock tests. The teacher checks what's working and adjusts.

Test set = the final board exam. Taken **once**, at the end, with unseen questions. The honest score.

Why split at all?

If we test a model on the examples it learned from, it can score 100% just by **memorising**, like a student who memorised the answer key. We only know it truly learned by checking on **examples it has never seen**.

Why about 80% for training?

MORE DATA FOR TRAINING →

the model learns patterns better. **But** less is left to check it, so the score is less trustworthy.

MORE DATA FOR CHECKING →

a more reliable score. **But** the model learned from fewer examples.

80/10/10 or 70/15/15 is a common starting point: **most** data to learn from, **enough** kept aside for a fair check. It's a habit, **not a rule**:

Dataset size	Typical split (train / val / test)	Why
Small (hundreds to a few thousand)	60 / 20 / 20, or cross-validation	Check sets need a bigger share to have enough examples.
Medium (tens of thousands)	70 / 15 / 15 or 80 / 10 / 10	The classic balance.
Huge (millions)	98 / 1 / 1	1% of 10 million is still 100,000 rows: plenty to check with.

Try it: split calculator



The split is random, so it uses a **seed / random_state** (section 8) to make it repeatable. For rare classes, make it a **stratified** split.

What exactly is the validation set for?

- **Which model?** Compare a simple and a complex model; keep whichever does better on validation.
- **Which settings?** Settings we choose *before* training (how complex the model is, how long to train) are called **hyperparameters**. Try a few and keep the best on validation.
- **When to stop training?** Watch the validation score and stop when it stops improving (section 10).

! COMMON MYTH: "VALIDATION IS ONLY FOR DEEP LEARNING"

Not true. Validation is used in classic machine learning too. The difference is *how*:

Where	How validation is used
Classic ML	To compare models and pick settings . With small data, often cross-validation : split training data into 5 parts, train on 4, check on the 5th, rotate 5 times and average.
Deep learning	Checked after every full pass through the training data (an epoch , explained in Session 2). Training stops when validation stops improving: early stopping .
Fine-tuning an LLM	Same idea: a held-out validation set is checked during fine-tuning so the model doesn't just memorise the small dataset.

! COMMON MYTH: "THE VALIDATION SET PREVENTS OVERFITTING"

It **detects** overfitting, like a smoke alarm. **We** then act on it (section 10).

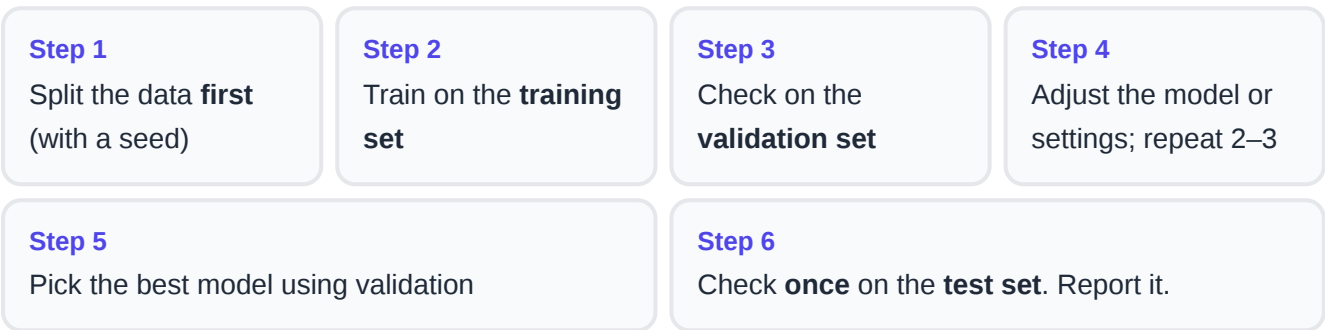
The test set: why only once, at the very end?

If we keep checking the test score and tweaking the model, we slowly **tune the model to the test set**. The final score becomes **too optimistic**, and the model does worse in the real world.

ANALOGY

A teacher who sees the board-exam questions and keeps adjusting lessons to match them: marks go up, real understanding doesn't.

The full workflow



Splitting mistakes to avoid

Mistake	Why it's a problem	Do this instead
Shuffling time-based data	The model "sees the future" during training	Train on the past, validate on later data, test on the most recent
Same person in train and test	The model recognises the patient, not the disease	Split by person / patient / customer, not by row
Duplicates across splits	The test set secretly contains training rows	Remove duplicates before splitting
Rare class missing from a split	With 1% fraud, a random split may put almost none in test	Use a stratified split
Cleaning using the whole dataset	Test-set information leaks into training	Split first, then decide cleaning steps on training data only
No seed	Every run gives a different split, so results can't be reproduced	Fix the random_state

Takeaway: **Train** to learn, **validate** to choose and tune (in classic ML *and* deep learning), **test** once for the honest score.

10. Did the model really learn? Underfitting, overfitting, bias and variance

The goal is **not** a perfect score on the training data. The goal is a model that does well on **new data it has never seen**. This is called **generalisation**.

THREE STUDENTS PREPARING FOR AN EXAM

- Student A** barely studied: does badly on practice papers *and* the real exam. → **Underfitting**
- Student B** memorised every past paper word for word: 100% on past papers, but fails when the exam asks the same idea in a new way. → **Overfitting**
- Student C** understood the concepts: does well on past papers *and* the new exam. → **Good fit (generalises)**

How to diagnose it: compare training and test scores

Example: a classification model, scored by accuracy (% of correct answers).

Case	Training accuracy	Test accuracy	Gap	Diagnosis
A	62%	60%	Small, but both low	Underfitting (high bias): hasn't learned the pattern even on data it has seen
B	99%	71%	Big gap	Overfitting (high variance): learned the training data "too well", including its noise, so it doesn't work on new data
C	91%	89%	Small, both high	✓ Good fit : learned the real pattern and generalises
D	100%	100%	None	Suspicious! Too good to be true. Check for data leakage (a test row or the answer hidden in the features).

RULE OF THUMB

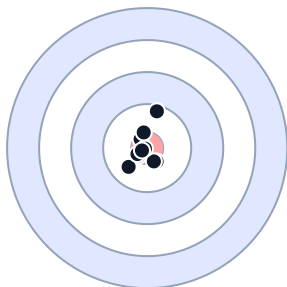
Both scores low → underfitting. **Training high, test much lower** → overfitting. **Both high and close** → good fit.
(During building, compare training with **validation**; the test set is saved for the final check.)

Bias and variance: the two kinds of mistakes

	Plain meaning	Shows up as	Goes with
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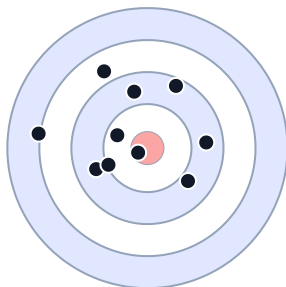
Bias	The model's assumptions are too simple , so it misses the real pattern in the same way every time. (Example: drawing a straight line through data that clearly curves.)	Low training score <i>and</i> low test score	Underfitting
Variance	The model is too sensitive to the exact training rows. Change the training data a little and the model changes a lot, because it learned the noise.	High training score, much lower test score	Overfitting

Picture 4 players throwing darts, where the bullseye is the true answer:



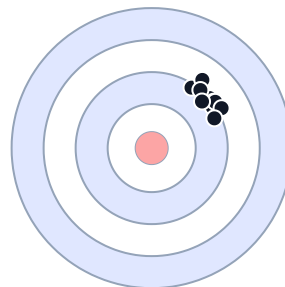
Low bias, low variance

✓ Good fit: consistently on target



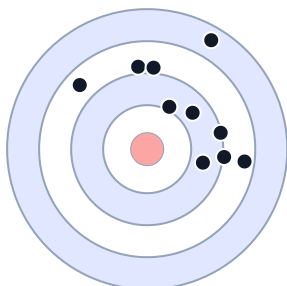
Low bias, high variance

Overfitting: right on average, but all over the place



High bias, low variance

Underfitting: consistent, but consistently wrong



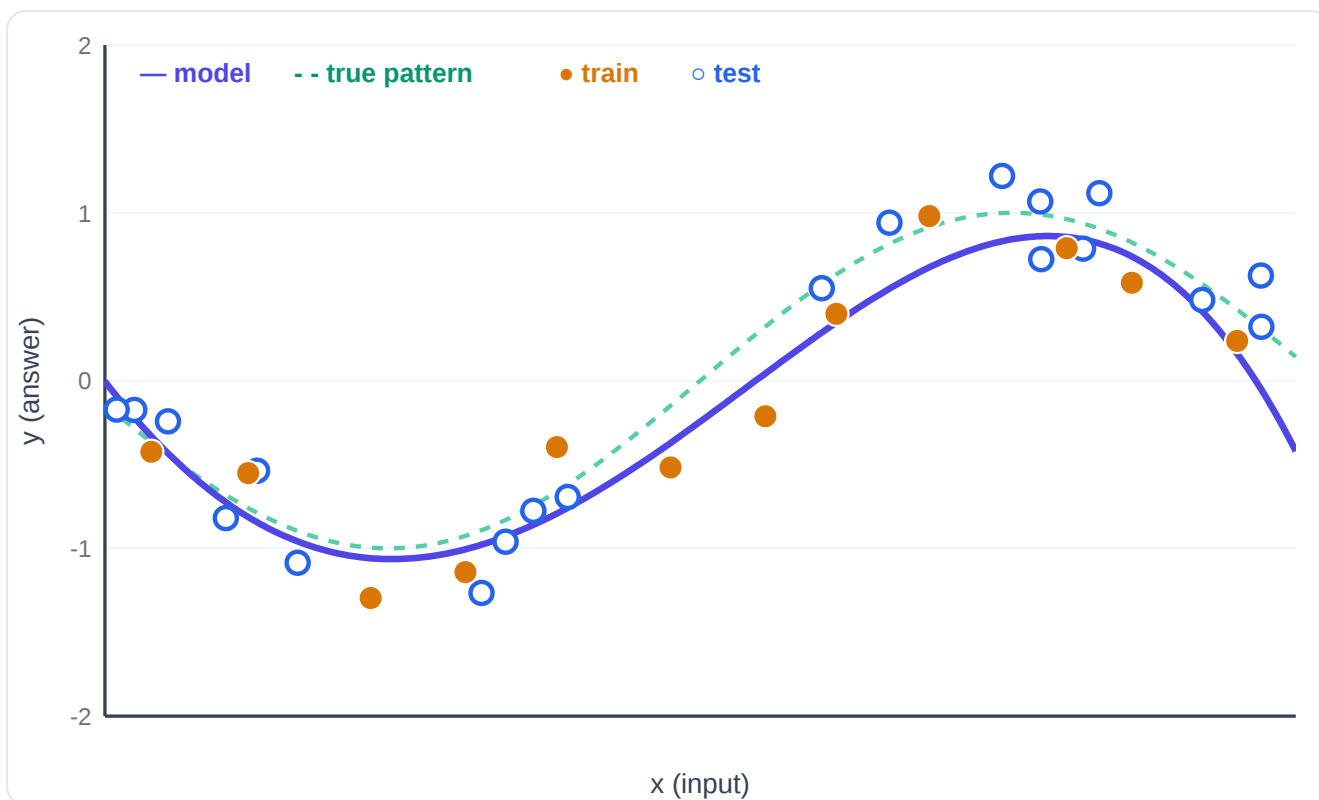
High bias, high variance

✗ Worst of both

THE TRADE-OFF

Making a model **more complex** lowers bias (it can follow the pattern better) but raises variance (it starts following the noise too). Making it **simpler** does the opposite. We look for the **middle**: complex enough to catch the pattern, simple enough to ignore the noise.

Try it: model complexity



Model flexibility: **4 (flexible curve)**

Training error: **0.044**

Test error: **0.070**

Lower is better. Watch the **gap** between them.

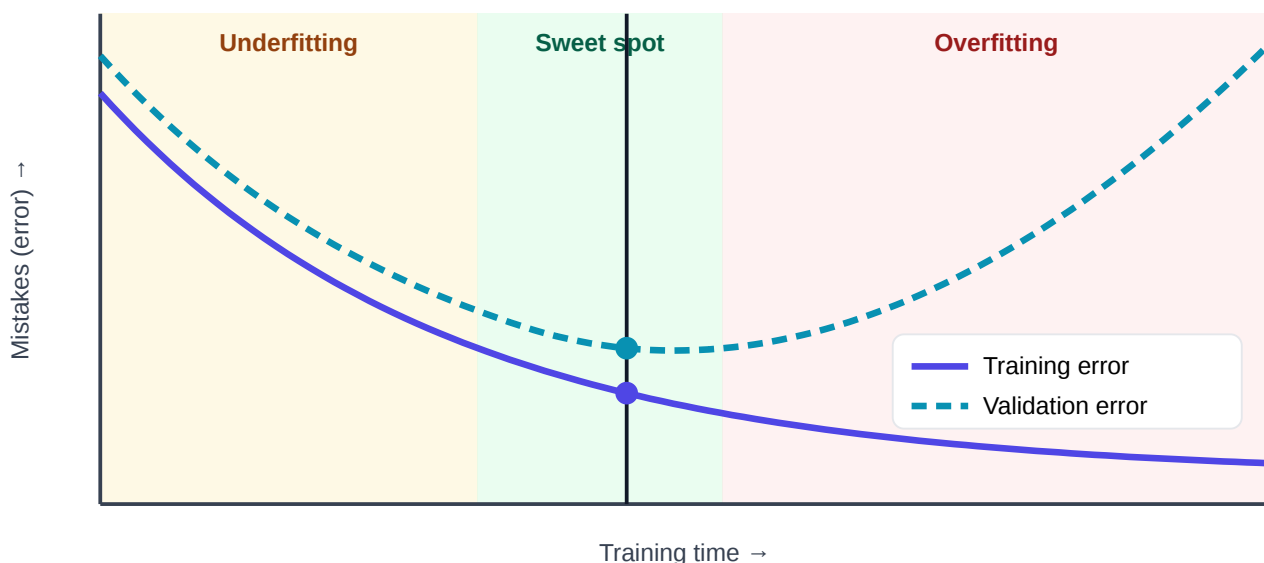
✓ **Good fit: follows the real pattern. Both errors are low and close together: it generalises.**

WHAT THE SLIDER SHOWS

Low flexibility (straight line): misses the curve, so both errors are high: underfitting. **Medium**: follows the real curve, and both errors are low: good fit. **High flexibility**: the curve wiggles through every training dot, so training error is almost 0 but test error shoots up: overfitting.

Watch it happen during training

The same story over time. Early in training the model hasn't learned enough; later it starts memorising. The **validation** error tells us when to stop.



Why it happens, and what to do

	Common reasons	What to try
Underfitting (high bias)	• Model too simple for the pattern (a straight line for a curve) • Features don't carry enough information (predicting house price from paint colour only) • Not trained long enough • Too much regularisation (Session 2)	• Use a more flexible model • Add better or more informative features • Train longer • Reduce regularisation
Overfitting (high variance)	• Model too complex for the amount of data • Too little training data • Noisy data or wrong labels (the model memorises the mistakes) • Trained for too long • Too many irrelevant features (ID numbers, house number...)	• Get more (and cleaner) training data • Use a simpler model • Add regularisation (L1 / L2, Session 2) • Early stopping, using the validation set • Remove irrelevant features • Use cross-validation to check the result is stable

What a good model looks like

A GENERALISED MODEL

- ✓ Good score on data it has **never seen** (validation and test)
- ✓ Training and test scores are **close** to each other (small gap)
- ✓ Learned the **real pattern**, not the noise or quirks of the training rows
- ✓ Its result stays similar with a different split or seed

WHY WE DON'T CHASE 100% ACCURACY

- **Real data has noise.** Labels contain mistakes, and some things are simply unpredictable (two identical-looking loan applicants can behave differently). No model can be right about pure chance.
- **100% on training data usually means memorising**, which is exactly overfitting.
- **100% on the test set** is more often a sign of **data leakage** than of a brilliant model.
- **"Good enough" depends on the problem.** 90% might be great for recommending movies and not enough for a medical test. Compare against a simple baseline (for example "always predict the most common class") and against how well humans do.

Takeaway: Underfitting (high bias) = too simple: bad on train *and* test. **Overfitting** (high variance) = memorised: great on train, poor on test. Aim for a **generalised** model: good and similar scores on data it has never seen, not 100%.

11. Choosing the right dataset for a problem

A simple model with the **right data** usually beats a fancy model with the **wrong data**.

The 5-step method

Step	Ask yourself	Example: "Detect fake product reviews"
1. Define the problem	What should the model do, in one sentence?	Say whether a review is genuine or fake.
2. Define the output	A number, a category, a group or new content?	Genuine / fake → binary classification
3. Define the inputs	What would a human expert look at?	Review text, rating, reviewer history, time posted
4. Check the labels	Do answers exist, or must someone create them? (No labels → consider unsupervised)	Need reviews already marked genuine or fake
5. Evaluate candidates	Run each dataset through the health check below	Public dataset vs. the company's own flagged reviews

Try it: problem → dataset finder

Predict the selling price of a flat

Output	A price in ₹
Task and learning type	Regression: continuous · supervised
Data type	Structured / tabular
Features (X)	Size, location, bedrooms, age, floor
Label (y)	Past selling prices
Watch out for	Outliers; old prices in a fast-changing market

Predict whether a customer will cancel their subscription

Output	Yes / no
Task and learning type	Classification: binary · supervised
Data type	Structured / tabular
Features (X)	Months subscribed, usage, complaints, plan type
Label (y)	Did they cancel? (past records)
Watch out for	Imbalance: use a stratified split

Sort incoming support tickets into categories

Output	One of Billing / Login / Bug / Other
Task and learning type	Classification: multiclass · supervised
Data type	Text
Features (X)	Ticket subject and description
Label (y)	Category agents assigned
Watch out for	Mis-categorised tickets; private details

Find natural topics in 50,000 support tickets (no categories yet)

Output	Groups of similar tickets
Task and learning type	Clustering · unsupervised
Data type	Text
Features (X)	Ticket text
Label (y)	None needed
Watch out for	Humans must read and name each group; choose a sensible number of groups

Group our customers into types for marketing

Output	Groups nobody has defined yet
Task and learning type	Clustering · unsupervised
Data type	Structured / tabular
Features (X)	Visits, spend, categories, location
Label (y)	None needed
Watch out for	Features on very different scales; groups must make business sense

Spot unusual card transactions (no fraud labels)

Output	Normal / unusual
Task and learning type	Anomaly detection · unsupervised
Data type	Structured / tabular
Features (X)	Amount, time, location, device, merchant
Label (y)	None (or a few confirmed cases to check with)
Watch out for	Unusual is not always fraud: humans review the flags

Predict next month's electricity demand

Output	A number over time
Task and learning type	Regression: forecasting · supervised
Data type	Time series
Features (X)	Past usage, temperature, holidays
Label (y)	Actual demand
Watch out for	Split by time, never shuffle

Build a chatbot that answers questions from our HR policy

Output	Generated text
Task and learning type	Generation · generative AI (LLM)
Data type	Text documents
Features (X)	HR policy documents
Label (y)	Usually none to start (RAG, later)
Watch out for	Outdated policy versions; confidential content

Dataset health check

- ☐ **Relevant:** Does it match my problem and where it will be used?
Red flag: US house prices for an Indian property app
- ☐ **Labelled correctly:** Are the answers present and trustworthy? (Or is this an unsupervised problem?)
Red flag: Labels created by guessing or a buggy script
- ☐ **Large enough:** Enough examples of every class, with some left for validation and test?
Red flag: Only 12 examples of the class you care about most
- ☐ **Representative:** Covers the situations the model will meet?
Red flag: Only daytime photos for a night-time camera
- ☐ **Balanced (or handled):** Is one class hugely dominating?
Red flag: 99.9% 'normal' with no plan
- ☐ **Clean:** Few missing, duplicate or wrong values?
Red flag: Half the rows have blank key columns
- ☐ **Current:** Recent enough for today?
Red flag: Pre-2020 data for post-2020 behaviour
- ☐ **Legal & ethical:** Allowed to use it? Private data protected?
Red flag: No licence, or personal data without consent

12. Session 1 cheat sheet

The one page to keep.

① Words

Term	Also called / meaning
Feature (X)	input, independent variable, predictor, attribute, column
Label (y)	target, dependent variable, output, ground truth, class (classification)
Row	record, sample, example, instance, observation
Seed / random_state	A fixed number that makes random sampling and splitting repeatable
Bias / variance	Too simple (underfitting) / too sensitive to the training rows (overfitting)
Generalisation	Doing well on new, unseen data: the real goal

② Match the problem to the data

If your problem is...	Task type	Learning type	Data you need	Practice dataset
"How much?" (any value)	Regression: continuous	Supervised	Examples + number labels	House prices (Kaggle)
"How many?" (a count)	Regression: count	Supervised	Examples + count labels	Bike-sharing demand (UCI)
"Yes or no?"	Binary classification	Supervised	Examples + 2 labels	Titanic survival (Kaggle)
"Which one of several?"	Multiclass classification	Supervised	Examples + 1 of many labels	MNIST digits, Iris flowers
"Which tags apply?"	Multi-label classification	Supervised	Examples + a set of tags	Movie-genre datasets
"What happens next over time?"	Forecasting	Supervised	Time-stamped history	Store sales, weather
"Which items are similar?"	Clustering	Unsupervised	Examples, no labels	Mall customers (Kaggle)
"Is this unusual?"	Anomaly detection	Often unsupervised	Mostly normal examples	Credit-card fraud (Kaggle)

"What goes together?"	Association	Unsupervised	Baskets / transactions	Grocery basket datasets
"Write / answer / summarise"	Generation	GenAI (LLM)	Documents, Q&A pairs, instructions	Hugging Face text datasets

③ Sampling and splitting rules of thumb

Situation	Do this
Always	Fix a seed / random_state so the split can be reproduced
Rare class (e.g. fraud)	Stratified split, so every set has the same share of the rare class
Imbalanced training data	Over- or under-sample the training set only , after splitting
Small / medium / huge dataset	60/20/20 or cross-validation / 70/15/15 or 80/10/10 / 98/1/1
Time-based data	Split by time (past → future), never shuffle
Many rows per person	Split by person, not by row

④ Reading train vs. test scores

Training	Test	Diagnosis	First thing to try
Low	Low	Underfitting (high bias)	More flexible model, better features
High	Much lower	Overfitting (high variance)	More data, simpler model, regularisation, early stopping
High	High, close	Good fit	Check once on the test set and ship it
100%	100%	Suspicious	Look for data leakage

⑤ Where to find datasets

Source	Best for
Kaggle Datasets	Beginner-friendly data with example notebooks
Hugging Face Datasets	Text, audio and image data for NLP and LLM work
UCI ML Repository	Classic, small, clean datasets for learning
Google Dataset Search	A search engine for datasets
data.gov.in	Indian government open data

⑥ Red flags

No description of how the data was collected · no licence · labels you can't verify · one class almost missing · test data mixed into training · personal data without consent · data much older than the problem · a score that looks too good to be true

⑦ Is there a standard industry cheat sheet?

There isn't one official version. Good datasets come with documentation that answers these questions: **dataset cards** on Hugging Face, the **data description tab** on Kaggle, and the well-known "**Datasheets for Datasets**" question list. If a dataset has no documentation at all, treat that as a red flag.

13. Common questions and quiz

Questions learners usually ask

▼ Is data science the same as AI?

No. They overlap through machine learning. Data science is about getting insights from data (including dashboards and statistics that aren't AI). AI is about making machines act smart (including rule-based systems that don't learn from data).

▼ Why so many names for the same thing (feature, independent variable, predictor...)?

They come from different fields: statistics says "independent / dependent variable", ML says "feature / label", databases say "column / record". They mean the same thing.

▼ Is "predicting a number" always regression?

Only if the number is an **amount**. If it's really a code or a name (PIN code, product ID), it's classification.

▼ If there are no labels, how does the machine know it's right?

It doesn't, in the same way. We check that the groups are tight and well separated, and, most importantly, that people who know the business find them meaningful and useful.

▼ How many clusters should we ask for?

There's no single right answer. Try a few numbers, look at how well separated the groups are, and pick the number that is clear and useful for the business.

▼ Why do people always use 42 as the seed?

It's a joke from the book *The Hitchhiker's Guide to the Galaxy*. Any number works; the point is to use the same one so results can be reproduced.

▼ Is a high training score good news?

Only if the validation/test score is also high and close to it. A high training score with a much lower test score is overfitting.

▼ Why not use 100% of the data for training?

Then we'd have nothing unseen to check with, and couldn't tell learning apart from memorising.

▼ Do we use a validation set in classic machine learning?

Yes: to compare models and pick settings, often through cross-validation. Deep learning also checks it after every epoch to decide when to stop.

▼ Can I use any data I find online?

No. Check the licence, the site's terms and privacy laws. Personal data needs special care and consent.

Quick quiz

1. A bank writes "IF amount > ₹5 lakh AND new device THEN block". Is this machine learning?

Yes

No

A human wrote the rule: rule-based AI, not ML.

2. In a salary dataset, "years of experience" is the...

Dependent variable

Independent variable

It's an input (feature). Salary is the dependent variable.

3. Routing an email to exactly one of 5 departments is...

Binary

Multiclass

Multi-label

Several options, exactly one answer.

4. You have 1 lakh customer records and no labels. You want to find customer types. This is...

Classification

Clustering

No labels + find groups = clustering (unsupervised).

5. You asked the machine for 5 clusters. Does that prove the data has 5 real groups?

Yes

No

It always makes as many groups as you ask for. Humans must check they mean something.

6. Your data has 2% fraud. Which sampling keeps 2% fraud in both train and test?

Simple random

Stratified

Convenience

Stratified sampling keeps each group's proportion.

7. Why fix a random seed (random_state) when splitting data?

It makes the model more accurate

So the same split can be reproduced by anyone

Same seed → same 'random' sample → reproducible results.

8. Training accuracy 98%, test accuracy 65%. Diagnosis?

Underfitting

Overfitting

Good fit

Big gap between training and test = overfitting (high variance).

9. Training accuracy 58%, test accuracy 56%. Diagnosis?

Underfitting

Overfitting

Good fit

Both low = underfitting (high bias): the model is too simple or the features are weak.

10. Which is a fix for overfitting?

Use a more complex model

Get more training data / add regularisation

More data, simpler model, regularisation or early stopping all fight overfitting.

11. Your model scores exactly 100% on the test set. First reaction?

Celebrate

Check for data leakage

Real data has noise; 100% is usually a sign that test information leaked into training.

12. You have 10 million rows. Which split makes most sense?

60/20/20

98/1/1

1% of 10 million is still 100,000 rows.

Coming up in Session 2: we know the model learns *from* data. Next: **how** does it learn? We'll build a linear regression model, measure its **loss**, and see how **gradient descent** updates its **weights** step by step. You'll also meet L1 / L2 regularisation, one of the fixes for overfitting from section 10.